

Lab-1: Continuous System Simulation – Chemical Reactions Simulation

Lab Date: December 17th, 2025

Deadline: December 24th, 2025

Class: CSIT 5th Semester, New Summit College

Subject: Simulation & Modeling

Submission Mode: **Handwritten Document** (Print Output Section Only)

Objective:

The primary objective of this lab is to simulate a continuous system representing chemical reactions in C programming. The students will implement a system where two chemicals react to produce a third one, analyze the system's behavior over time, and visualize the results using scatter plots.

Learning Outcomes

After completing this lab, students will be able to:

- Understand the dynamics of continuous systems and how variables change over time.
- Write C programs to simulate and solve differential equations.
- Analyze and interpret simulation data to understand the kinetics of chemical reactions.
- Visualize results using scatter plots to represent changes in chemical concentrations.

Tasks:

1. Task 1: Coding the Reaction Simulation

- a. Write a program to implement the above differential equations using a discrete time step (Δt).
- b. Use the provided pseudo-code or implement using arrays to store concentrations at each time step.

- c. Ensure proper file handling to save the output results for visualization.

2. Task 2: Simulate with Given Parameters

- a. Use the given initial values and parameters.
- b. Simulate the system for 100 seconds.

3. Task 3: Plot the Results

- a. Generate a scatter plot using software like MATLAB, Python (Matplotlib), or Excel to visualize $C_1(t)$, $C_2(t)$, and $C_3(t)$.
- b. Label the axes and provide a legend for clarity.

Lab Context:

Let us consider C_1 , C_2 , and C_3 as the amounts of Ch_1 , Ch_2 , and Ch_3 at any instant of time t . The rate of increase of C_1 , C_2 , and C_3 are given by the following differential equations:

$$\frac{dC_1}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_2}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_3}{dt} = 2K_1C_1C_2 - 2K_2C_3$$

Where K_1 and K_2 are constants.

As soon as the chemicals Ch_1 and Ch_2 are mixed, the reaction starts, and the amounts of C_1 , C_2 , and C_3 in the mixture change over time. The simulation of the reaction will determine the state of the system, i.e., the values of C_1 , C_2 , and C_3 at different points in time.

Report Structure:

- **Objective:** Clearly state the objective of the lab.
- **Theory:** Include a brief explanation of:
 - a. Continuous system and

b. Chemical reaction kinetics.

- **Procedure:** Outline the steps to simulate the system in C and visualize the results.
- **Code:** Include the complete program code.
- **Results:** Provide the scatter plot and discuss the observed trends.
- **Conclusion:** Summarize the learnings and observations from the simulation.

References:

1. <https://colab.research.google.com/drive/1AUge3bU-dsIP-75n2tCXaya3QcbFOe3?usp=sharing>
2. <https://drive.google.com/file/d/1WlpP5KmuVOaQQx570G0Fsd3iFwpBesRr/view?usp=sharing>